BOUNDED COHOMOLOGICAL DIMENSION AND COMMUTATION WITH DIRECT SUMS MAKE THE EXTENSION BY ZERO OF THE STRUCTURE SHEAF COMPACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to the derived global sections, the use of the two hypotheses in the form of finite cohomological dimension and acyclicity of direct sums of injectives, the choice of K-injective representatives, and the computation on the resulting complex of acyclics.

Editorial note. The label lemma-when-jshriek-compact names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Compactness of j_!O_U and the identification of its Hom functor. Same chapter, the identification Hom(j_!O_U, K) = R Gamma(U, K) and the notion of a compact object of the derived category. Supplies the reduction of the compactness statement to the commutation of R Gamma(U, -) with direct sums.
- K-injective resolutions and the computation of right derived functors on acyclics. Injectives, the theorem providing a K-injective embedding, the lemma that direct sums of K-injectives represent direct sums, and Derived Categories, the lemma computing an unbounded right derived functor on a complex of acyclics. Supplies the representatives used to compute R Gamma(U, direct sum of K_i).

Source reference. Original lemma 07K7: sources/stacks-chow/tags.txt, source lines 9647–9647. Prerequisite pointer invoked at proof line(s) 14158; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 07D9: sources/stacks-chow/tags.txt, source lines 9404–9404. Prerequisite pointer invoked at proof line(s) 14160; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 079P: sources/stacks-chow/tags.txt, source lines 9279–9279. Prerequisite pointer invoked at proof line(s) 14156; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let U be an object of $\mathcal{C}$. The $\mathcal{O}$-module $j_!\mathcal{O}_U$ is a compact object of $D(\mathcal{O})$ if there exists an integer d such that*

(1) *$H^p(U, \mathcal{F}) = 0$ for all $p > d$, and*
(2) *the functors $\mathcal{F} \mapsto H^p(U, \mathcal{F})$ commute with direct sums.*

Proof. Assume (1) and (2). Recall that $\mathrm{Hom}(j_!\mathcal{O}_U, K) = R\Gamma(U, K)$ for K in $D(\mathcal{O})$. Thus we have to show that $R\Gamma(U, -)$ commutes with direct sums. The first assumption means that the functor $F = H^0(U, -)$ has finite cohomological dimension. Moreover, the second assumption implies any direct sum of injective modules is acyclic for F. Let K_i be a family of objects of $D(\mathcal{O})$. Choose K-injective representatives $I_i^\bullet$ with injective terms representing K_i, see Injectives, Theorem 079P. Since we may compute RF by applying F to any complex of acyclics (Derived Categories, Lemma 07K7) and since $\bigoplus K_i$ is represented by $\bigoplus I_i^\bullet$ (Injectives, Lemma 07D9) we conclude that $R\Gamma(U, \bigoplus K_i)$ is represented by $\bigoplus H^0(U, I_i^\bullet)$. Hence $R\Gamma(U, -)$ commutes with direct sums as desired. □